Cyclic codes: are special linear block codes with an extra property

if a codeword is cyclically shifted, the result is another codeword.

ex- 1011000 → Code word

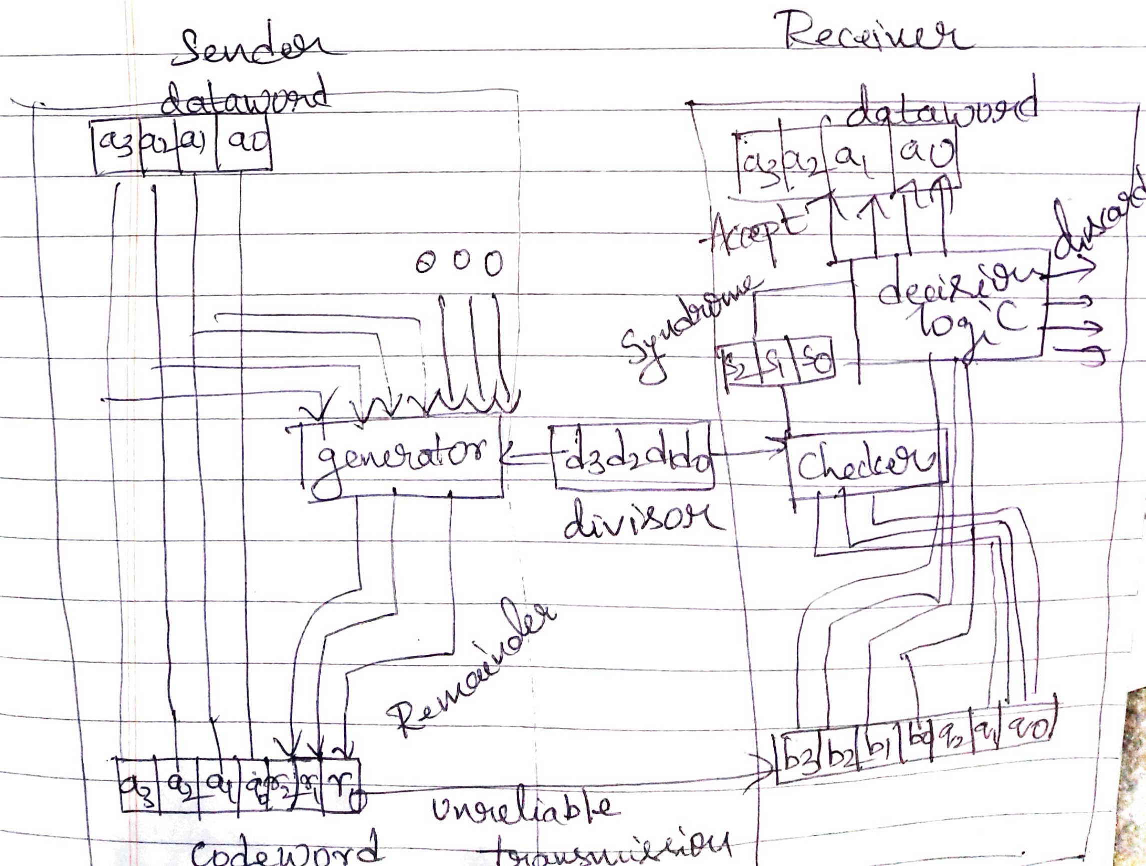
left-shift 0110001 — is a new word

Cyclic Redundancy check:

→ Error detecting code

→ Whenever a msg is transmitted, it may get scrambled by noise or data may get corrupted. To avoid this, we use error detecting codes, which are additional data added to a given msg which help us to find if any error occured in the data during the transmission.

CRC encoder & decoder



parties agree upon the size of the msg
generator polynomial G(x)

r is the Order of the G(x), r bits
appended to the lower end of M(x)

block is now divided by G(x) Using
ulo 2 division.

remainder after division is added to
x) & transmitted

1- The receiver receives data block &
receiver divides the block by G(x)

remainder is 'zero' the receiver
s the frame else rejects the frame.

dataword = 1001

divisor = 1011
$x^3 x^2 x^1 x^0$

```
        101          Quotient
     ________
1011 ) 1001000
divisor 1011
       -----
       00100
        0000
       -----
        01000
         1011
        -----
         00110
         remainder
```

0	0	0
0	1	1
1	0	1
1	1	0

Codeword = $\boxed{1001|110}$
data Remainder

Codeword received 1001110

```
        101
     _________
1011 ) 1001110
       1011
       ------
       001011
          1011
        ------
         00000
```

dataword accepted 1001

code word 1000|110

```
        1011
     __________
1011 ) 1000110
       1011
       ------
       001111
         1011
        -----
         01000
          1011
         -----
          0011
```

non zero

dataword discarded

Using polynomials

→ polynomial — binary word

1 0 0 0 1 1

x^5 x^4 x^3 x^2 x^1 x^0

dataword = x^3+1 1001

divisor = x^3+x+1 1011

1011000

x^0

6 5 4 3 2 1

x^3+x

$$
\begin{array}{r}
x^3+x \\
x^3+x+1\,\big)\ x^6 + x^3 \\
x^6+x^4+x^3 \\
\hline
x^4 \\
x^4+x^2+x \\
\hline
x^2+x
\end{array}
$$

remainder

1001000

6 5 4 3 2 1 0

x^6+x^3

$\boxed{x^6+x^3}\boxed{x^2+x}$

1

$$\boxed{\begin{array}{l} \text{Data} = x^5 + x^2 = 100100 \\ \text{polynomial} = x^3 + x^2 + 1 = 1101 \end{array}}$$

degree = 3

$100100\overline{000} = \boxed{x^8 + x^5}$

$x^8 \quad x^7 \quad x^6 \quad x^5 \quad x^4 \quad x^3 \quad x^2 \quad x^1 \quad x^0$

$$x^3+x^2+1 \,\overline{)\, x^8 + x^5 \,}\, (x^5+x^4+x^3+x^2+1$$

$- \quad \cancel{x^8} + x^7 + \cancel{x^5}$

$\cancel{x^7}$

$\cancel{x^7} + x^6 + x^4$

$\cancel{x^6} + x^4$

$- \quad \cancel{x^6} + x^5 + x^3$

$\cancel{x^5} + \cancel{x^4} + x^3$

$\cancel{x^5} + \cancel{x^4} + x^2$

$\cancel{x^3} + \cancel{x^2}$

$\cancel{x^3} + \cancel{x^2} + 1$

1

3 bits — 001

$100100\boxed{001} = \boxed{x^8 + x^5 + 1}$

8 7 6 5 4 3 2 1 x^0

Reciever knows = $\boxed{x^3 + x^2 + 1}$

bitstream = 10011101

polynomial generator = x^3+1 = 1001

(10001100

```
1001 ) 100111101000
       1001↓↓↓↓
       00001101
           1001
            1000
            1001
            000100
              0000
               100   remainder
```

$\boxed{\text{Data} = x^5 + x^2}$

= 100100

$$\begin{array}{cccccc} 1 & 0 & 0 & 1 & 0 & 0 \\ x^5 & x^4 & x^3 & x^2 & x^1 & x^0 \end{array} = 100100$$

data

$\boxed{\text{Polynomial} = \underset{x^3\,x^2\,x^1\,x^0}{1101} = x^3+x^2+1}$

Order = 3

100100000

(111101

```
1101 ) 100100000
       1101↓
       01000
        1101
        01010
         1101
         01110
          1101
          001100
             1101
             0001   remainder
```

x^8+x^5+1

100100|001

x^8 ... 0

$$\text{Data} = x^3+1 = 1001$$
$$\text{divisor} = x^3+x+1 = 1011$$

~~1011~~ 000

x^0 x^1 x^2 x^3 x^4 x^5 x^6

$\cancel{x^6}+$

1001 000

6 5 4 3 2 1 0

$$\boxed{x^6+x^3}$$

$$
\begin{array}{r}
x^3+x+1 \,\big)\, \cancel{x^6}+\cancel{x^3} \quad (x^3+x \\
\cancel{x^6}+\cancel{x^3}+x^4 \\
\hline
\cancel{x^4} \\
\cancel{x^4}+x^2+x \\
\hline
x^2+x
\end{array}
$$

dataword + Remainder

x^3+1+x^2+x

$$\boxed{\begin{array}{l} \text{data} = 1001 \\ \qquad\quad\; 1011 \end{array}}$$

$1001\,\underline{000}$

1011

$$
\begin{array}{r}
1011 \,\big)\, 1001000 \quad (1010 \\
1011 \\
\hline
001000 \\
1011 \\
\hline
00110 \\
0000 \\
\hline
110
\end{array}
$$

remainder

$$
\boxed{\begin{array}{r}
1011 \,\big)\, 1001000 \quad (1010 \\
1011 \\
\hline
00100 \\
0000 \\
\hline
1000 \\
1011 \\
\hline
00110 \\
0000
\end{array}}
$$